

OVERVIEW



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Wilding the predictive brain

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Abstract

The Predictive Processing (PP) framework casts the brain as a probabilistic prediction engine that continually generates predictions of the causal structure of the world in order to construct for itself, from the top down, incoming sensory signals. Conceiving of the brain in this way has yielded incredible explanatory power, offering what many believe to be our first glimpse at a unified theory of the mind. In this paper, the picture of the mind brought into view by predictive processing theories is shown to be embodied, deeply affective and nicely poised for cognitive extension. We begin by giving an overview of the main themes of the framework, and situating this approach within embodied cognitive science. We show perception, action, homeostatic regulation and emotion to be underpinned by the very same predictive machinery. We conclude by showing how predictive minds will increasingly be understood as deeply interwoven with, and perhaps extended into, the surrounding social, cultural and technological landscape.

This article is categorized under:

Philosophy > Foundations of Cognitive Science

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1 | INTRODUCTION

Recently a new perspective dominates many discussions of mind and cognition. That perspective depicts the brain as essentially a probabilistic prediction engine, dedicated to the task of minimizing the disparity between how it expects (predicts) the world to be and the evidence presented by the sensory flow.¹ Part of the power of the framework lies in the elegant suggestion that much of what we take to be central to human intelligence—perception, action, attention, emotion, learning, and language—can be understood within a simple framework of predictions and error reduction. In what follows we will refer to this general approach to understanding the mind and brain as *predictive processing* (PP).

While the predictive processing framework is in many ways revolutionary, it can appear to commit proponents to a traditional “neurocentric” stance concerning the mind. Such a stance depicts the mind as, in essence, what brains do,

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TABLE 1 Evolution of the predictive processing framework

Evolution of the framework	Key readings
Perceptual inference	Helmholtz, 1867; Lee & Mumford, 2003; Rao & Ballard, 1999
Simple active inference	Clark, 2013, 2016; Friston, Adams et al 2012; Friston et al. 2015, 2016, 2017
Embodied active inference	Allen & Friston, 2018; Allen & Tsakiris, 2019; Barrett & Simmons, 2015; Bruineberg, Kiverstein, & Rietveld, 2018; Gallagher, 2017; Gallagher & Allen, 2018; Hesp et al., 2019; Joffily & Coricelli, 2013; Kirchhoff, 2018; Kiverstein, Miller, & Rietveld, 2017; Linson et al., 2019; Bruineberg & Rietveld, 2014; Seth, 2013; Van de Cruys, 2017
Extended predictive minds	Constant et al., 2019; Kirchhoff & Kiverstein, 2019

and it has tended to downplay the contributions of body, world, and action. However, a closer look reveals a much richer picture. Today, many researchers are exploring views of the predictive brain that allow for the body and the surrounding environment to make robust contributions to the predictive process itself. These recent developments strongly suggest that while it is true that predictive models can get us a long way in making sense of what drives the neural-economy, a complete picture of human intelligence requires us also to explore the many ways that a predictive brain is embodied in a living body, permeated with affect and embedded in an empowering socio-cultural niche. In this paper, we explore this promising evolution (see Table 1).

We start by introducing the predictive processing framework itself. We go on to outline how different camps within predictive processing research are viewing its amenability to embodied cognitive science, and lay some conceptual groundwork for ways to think about the predictive mind as active and embodied. We then turn to recent work on feelings and emotions, and review a new take on affect value and its role in directing core estimations of the reliability and value of prediction errors. We conclude by showing how such an active, embodied and affect-laden predictive organization would be well-poised to extend cognitive processing into the social, cultural and technological landscape.

2 | PERCEPTUAL INFERENCE

An increasingly popular theory in philosophy and neuroscience is that the perceiving brain is fundamentally an engine of prediction (Bubic, Von Cramon, & Schubotz, 2010; Clark, 2013; Friston, 2010; Hohwy, 2013). This engine, so the story goes, makes sense of the world by actively generating predictions of the incoming sensory stream. According to this view, much of what we take to be central to human intelligence—perception, action, attention, cognition, emotion, learning and language—are all underpinned by this common predictive mechanism (Clark, 2013, p. 181). In this paper, we assume that some version of the predictive processing story has merit—for a balanced review of the state of the evidence, see Walsh, McGovern, Clark, and O'Connell (2020).

This bedrock “predictive processing” (PP) story has roots in influential accounts of perception (Gregory, 1997; Helmholtz, 1867; Kant, 1781). How does the brain unearth, and enable us to perceive, a complex world of objects and events given the many challenges posed by noise and ambiguity? This challenge has been referred to as the “problem of perception” (Hohwy, 2013). To bring this into focus, consider the fact that any object can induce a vast variety of sensory patterns: the same object can be encountered from different angles or in changing atmospheric conditions. Moreover, many different objects produce similar sensory signals: a picture of an object and the object itself, or a partially obscured object and a fragmented one (e.g., a cat walking behind a picket fence). To make matters worse, we must also be able to explain how the brain is able to separate out the salient information from the unimportant sensory noise. The PP model “earns its salt” by offering an elegant and unified solution to such challenges.

In extremely brief terms, the predictive brain gets a grip on the noisy, ambiguous world presented by the sensory flow by continually learning about the vast network of temporal-spatial regularities that reflect, and in some ways constitute, its environment. Our world is filled with learnable regularities—both natural and synthetic, ranging from the very fast (e.g., patterns in a swiftly moving river) to very slow (e.g., the river eroding the land around it). Through tracking fast regularities we grasp the fine-grained details about a scene. Tracking slower regularities provide the kind of abstracted and enduring information (e.g., about the unchanging identity of the river) that can help contextualize and constrain more local predictions, and informs predictions of how states of affairs are likely to evolve on longer timescales.

The predictive brain uses its knowledge of regularities and patterns to make increasingly refined predictions about what objects and events are most likely to be responsible for the signals it receives from the environment. Perceptual experience, then, is the top-down “best guess” at the hidden causes of incoming sensory signals. Where there is a significant mismatch between prediction and incoming signals, the discrepancy (the residual error, or “prediction error”) moves forward (or “up”) through the hierarchical system helping to refine predictions or (see next section) to recruit actions aimed at making the sensory stream fit proprioceptive predictions, or at improving the state of information.

The prediction error minimizing routine is modulated at every level by a set of second-order expectations that track the reliability, or inverse variance, of the predictive system's own estimates given the state of the organism and the current context. It uses this estimation of reliability (referred to as “precision weighting”) to flexibly adjust the gain (like turning the volume up) on particular error units. Error units carry all the unexplained sensory information, and increasing their weighting increases the impact that information has on the unfolding process. This allows the system to flexibly modify the degree to which it relies on incoming prediction errors from the sensory periphery or prior beliefs about the state of the world (see Friston, 2009, 2010). For example, while listening to your favorite song in the shower, it would be useful to turn down the influence on the sensory signals produced by the flowing water and rely more on our clear memories of the song (Clark, 2016, p. 92).

The novel addition this theory makes to traditional, feedforward-dominated perception research is that perception is not explained by incoming signals alone, but crucially also includes active top-down predictions about their shape, reliability and what they could mean. Perception thus constructed is sometimes said to involve a kind of “controlled hallucination”—what we see is impacted by what we predict ought to be out there. Of course what we predict is itself continually tuned by the actual sensory signals, which works (in normal functioning brains) to anchor those predictions to reality. Nonetheless, the hard perceptual work is here accomplished mostly by the internal model (the “generative model”) that is constructing the predictions, leaving the incoming sensory information the task of (in effect) critiquing those predictions until a better fit is achieved.

For example, Anthony Norcia's Coffin Illusion (Figure 1) is usually first perceived as a grid of squares, like wooden panels on a door. However, once prompted to search for the circles, our subpersonal predictions about the content of the image, and so the corresponding experience of it, shifts dramatically. This despite the fact that the incoming pattern of sensory stimulation has not altered at all.

This approach openly opposes classical feedforward-dominated perceptual models. In the not too distant past the brain was commonly characterized as a relatively passive organ. Dormant neurons were thought to patiently await incoming signals to jolt them into action. When signals did arrive, they were thought to roll in from the sensorium and flow upward through the neural hierarchy increasing in complexity along the way (Marr, 1982). In direct contrast, PP

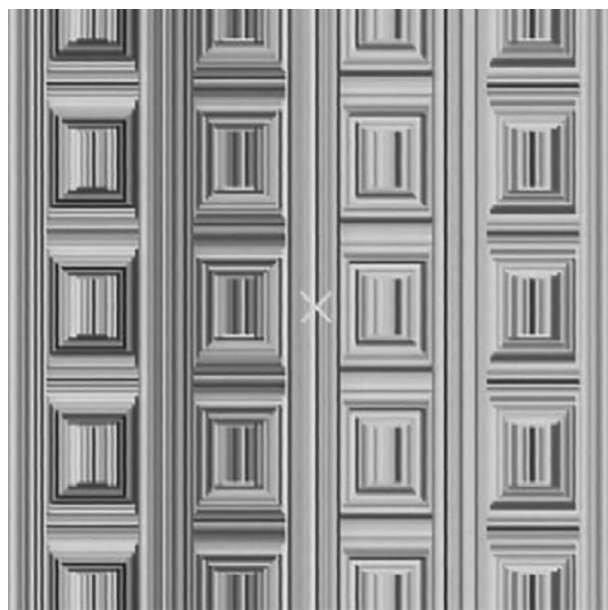


FIGURE 1 The Coffin Illusion (Norcia, 2006)—see separate file. Image is licensed under Creative Commons (CC BY-NC-SA 4.0)

describes the brain as fundamentally proactive—the brain actively generates perceptions by continually attempting to recreate from the top-down the world of sensory signals.

3 | SIMPLE ACTIVE INFERENCE

If we think primarily about perception alone, it can seem as if the fundamental goal of the predictive brain is to reconstruct the distal environment on the basis of noisy and ambiguous sensory information. But as noted above, PP places action at the heart of the prediction error minimizing process too. This opens the door to a non-reconstructivist understanding that is more closely aligned with work on the embodied mind (Chemero, 2011; Varela, Thompson, & Rosch, 2016). Such approaches reject the characterization of our perceptual goals as reconstructive in the first place. As a classic statement of the tradition, cited by Clark (2016) puts it:

“The overall concern of an enactive approach to perception is not to determine how some perceiver-independent world is to be recovered; it is, rather, to determine the common principles or lawful linkages between sensory and motor systems that explain how action can be perceptually-guided in a perceiver-dependent world.” (Varela, 1991, p. 173)

Perhaps the active PP system can make do with something other than a faithful reconstruction of the distal environment? To take a well-worn but still useful example, Phillip Fink, Foo, and Warren (2009) show that a baseball outfielder need not first model the entire onward trajectory of the baseball relative to their position, and to the field, in order to then begin the act of moving to catch it. All that is needed is an ongoing coordination strategy called “Optical Acceleration Cancellation” (OAC)—that involves moving such that the ball stays at a stable position in the retinal field, until it is close enough to catch. In order to successfully execute this strategy an agent requires no internal physics engine, no knowledge of aerodynamic equations governing the flight of a sound, slightly irregular, projectile in a mild North-Westerly wind. All they require is an understanding of the lawlike relations between their motor output and the position of the ball's projection on their retina.

In PP terms, as Clark (2016) explains, this becomes a matter of assigning high-precision weighting to errors related to the prediction that the optical projection of the ball remains at a stable location on the retina. In such a way the rest of the system's actions are recruited around the quashing of this particular error signal, to the neglect of most else happening on the field, until the desired state of catching the ball (or the undesired state of colliding with a teammate employing the same strategy) is reached. Here there is no prior process of tinkering at the generative model until a lack of overall error provides adequate comfort that we have formed an accurate representation of the external world, and action may now begin. Rather, successful action is itself the ongoing control of a small portion of the sensory flux within those constraints that the system predicts will lead toward its target state.

As Anil Seth (2014) suggests, we can think of this “non-reconstructivist” approach to PP as offering a mechanistic rendition of earlier sensorimotor theories of perception (O'Regan & Noë, 2001). Such “fast and frugal” strategies are much more suited toward the ongoing guidance of an organism that must constantly keep afloat in a fast-changing environment (Clark, 2016). For Clark, the availability of locally-effective non-representational strategies is not an argument that we should abandon all representation talk however. Rather the strength of active PP is the offer of, “a systematic way of combining deep, model-based flexibility with the use of multiple, fast, efficient, environmentally exploitative, routes to action and response” (2015, p18). In order for the PP system to effectively deploy such “fast and frugal” strategies as OAC, it must also be able to monitor slower-changing contextual factors (such as whether one is actually engaged in a game of baseball, or merely a participant) in order to ascertain when the circumstances are ripe for their deployment. This is why the PP system requires hierarchical depth, such that high-level states may target these large-scale increasingly invariant patterns throughout the fast fluctuations of the sensory stream.

Unlike on the reconstructivist story, these high-level action-oriented representations do not allow us to “throw away the world” (Clark, 1999) when we engage in planning our next action, but rather coordinate our interactions with the world at multiple levels of spatiotemporal grain. Nor is the correctness of these action-oriented representations contingent upon the rejection of any skeptical hypotheses. If a current affordance for ball-catching-action is correctly detected, then deploying OAC will guide the evolution of the skilled outfielder's sensorimotor interactions to the target ball-in-hand state. This model of current sensorimotor contingencies fulfills its purpose successfully—and it does so, we should

also note, irrespective of whether the hidden causes interacting with our sensorimotor array were instantiated by mischievous demons, curious scientists, or strange and charming fundamental particles.

4 | EMBODIED ACTIVE INFERENCE

If not the goal of representational fidelity, then what is the end toward which our sensorimotor interactions are being coordinated? Within the recent literature on PP, we can distinguish another (even more radically embodied) strand that treats the active, embodied brain in ecological and enactive terms (Allen & Friston, 2018; Bruineberg et al., 2018; Gallagher & Allen, 2018; Kirchhoff, 2018).

Enactivists subscribe to a mind-life continuity thesis that takes cognitive processes and living processes to work according to the same fundamental organizing principles with the shared imperative of maintaining the organism's integrity through adaptive regulation of environmental interactions (Di Paolo, 2005; Thompson, 2007; Varela, 1991). In working to produce, sustain and conserve its identity over time in its interactions with the environment, the organism enacts or “brings forth” a meaningful world, a process termed “sense-making”. Cognitive and affective processes work together in this framework to steer the organism through the world in pursuit of what is significant. Colombetti writes that “Cognition from an enactive perspective is, rather, the capacity to enact or bring forth a world of sense, namely, an Umwelt that has a special significance for the organism enacting it... cognition as sense making entails that cognition is simultaneously also affective.” (Colombetti, 2014, p. 18).

In this ecological-enactive rendition of PP, affect and cognition unfold in ways that are deeply reciprocally interactive with the whole animal-environment system. Environmental surroundings, meanwhile, are conceived in relation to the affordances or action possibilities they offer to the organism. A familiar illustration that effectively captures this logic in its simplest form is that of bacterial chemotaxis. As Varela (1991) describes, sugar is necessary to fuel the metabolic processes responsible for the ongoing production of the bacterium's body, thus sugar gains affective significance in relation to the bacterium. And because sugar is not, typically, evenly distributed throughout a solution, so a bacterium must engage in adaptive actions in order to seek out and move toward increasing concentrations of it. If it does not, it will not remain a bacterium much longer. The nutritional value of sugar and the imperative of moving toward it has no pre-existing validity independent of the bacterium, but rather is brought about by the relation between its needs and activities, and the sugar-seeking affordances of its environment.

As we have noted, cognition is typically separated from affect in the philosophical and psychological literature on emotion. Appraisal and evaluation are associated with cognition, while affect is identified with changes in the autonomic system in the body. Appraisal according to this ecological-enactive approach is instead understood as involving the whole living body of the organism (as it prepares to act on relevant affordances). Enactivists theorists of PP sometimes refer to this using the French-phenomenologist Maurice Merleau-Ponty's notion of a “tendency towards an optimal grip” (Bruineberg & Rietveld, 2014). “Grip” here refers to the organism's bodily stance in relation to its current situation. We use this term because grip, understood as bodily readiness for apt action, is something the organism must actively maintain in relationship with the changing environment. Prediction errors then signal an increase in disequilibrium in the organism-environment system as a whole—for instance, declining sugar levels, which will negatively impact readiness for action. Such disequilibria reflects a divergence in the sensory states (exteroceptive and interoceptive) the organism expects to occupy given the kind of organism it is, its current state and the niche it inhabits. The organism then acts to reduce this disequilibrium or to make it the case that it comes to occupy the sensory states that it expects to be in (given the life it leads). In other words, it acts so as to stay within the window of viability that defines it as the kind of being it is.

In this process, gross action is simply one component. Of equal importance is what might be thought of as “inner action”. Indeed, it has long been argued that the reason for having a brain is to keep organisms in “continuous equilibrium” with their environment (Pavlov, 1927), and as part of this process of efficient management of metabolic resources to drive survival and reproduction has been described as the “core task” of all brains (Sterling & Laughlin, 2015). In order to stay alive and viable, an organism needs to keep its bodily states within certain bounds. For instance, a body temperature of 40° centigrade is not conducive to continued existence for a human being. Regulating these “essential variables”—homeostasis—requires sensing the global physiological conditions of the body through afferent signaling to the brain of the internal state of one's body. Sensing of the global physiological conditions of the body—such as cardiac signals, states of the gut and viscera, air supply and glucose plasma levels—is known as interoception (Craig, 2003). Building a predictive model of the state of the body, via the very same inferential mechanisms underpinning perceptual

inference, allows the brain to engage autonomic action to bring the body into homeostatic balance. For instance, a hyperthermic animal can bring its body temperature into viable bounds through engaging autonomic reflexes such as perspiration. In order to stay viable on longer timescales, however, it will need to act—for instance it could move to a cooler place, like a shaded area under a tree. These actions—those that allow the brain to regulate the state of the body—are called allostasis. To stay viable on longer timescales organisms need to take prospective actions, anticipating dyshomeostatic conditions before they arise and acting to avoid them.

Enactivist predictive processing regards this fundamental imperative toward continued existence via homeostasis and allostasis a kind of “first prior” (Allen & Tsakiris, 2018). In other words, “[t]he brain is in the game of predicting the world, but only as a means to the end of embodied self-preservation” (Allen & Friston, 2018, p. 12). This means that the interoceptive signals, in tracking key homeostatic variables, are deemed highly precise—they are given “a priori hyper-precision of visceral channels” (Allen & Tsakiris, 2018). Unlike in perceptual inference, where beliefs are hypotheses can be much more malleably shaped so as to fit the world, interoceptive inference tracks physiological variables which must be maintained within quite narrow bounds (Seth & Friston, 2016). This is intuitive—the system that simply updates its perception of its body temperature to 50° centigrade, rather than acting to cool down (either through autonomic reflexes such as by perspiring, or allostatically by moving to a colder place) is unlikely to stay alive for long.

These visceromotor channels, then, are highly precise and resist simple revision in favor of action—that is, they minimize prediction errors given evolutionarily endowed prior expectations that the organism will continue to exist. But complex adaptive strategies also require prospective control, that seeks to minimize whole trajectories of error that reach far into the future. This demands deep *temporal models*, and recasts planning as a problem of probabilistic inference (Botvinick & Toussaint, 2012). On this view, action selection is based on prior beliefs about the expected consequences of a course of action or sequence of actions, referred to in the literature as an action “policy” (Friston et al., 2015, 2017). Expected prediction error is the error the agent predicts were it to pursue a particular action policy, given knowledge of the contingencies in the world, and knowledge of itself (such as the metabolic resources it has available to complete the action).

Intuitively, some actions are more likely than others to lead to desirable or “expected” outcomes—such as leaving a building through a ground floor door as opposed to a sixth floor window. Selection of actions that minimize expected error over longer timescales, thereby maximizing the probability of long-term existence here rests on a balance between the pragmatic and epistemic affordances of action. The pragmatic affordances of action—such as having a drink on a hot day—are readily apparent in terms of the drive to maintain homeostasis. Epistemic actions, by contrast, refer to actions that improve our state of information (Kirsh & Maglio, 1994; Kiverstein et al., 2017). It is here that curiosity and exploratory behavior find a home within an active inference framework (Friston et al., 2017; Kaplan & Friston, 2018; Kiverstein et al., 2017) as they enable organisms to increase their predictive grip in the long-term, by improving information and reducing key uncertainties (Friston et al. 2015, 2016, 2017). We return to these issues when we look at “extended predictive minds” in Section 5 below.

5 | EMOTIONS, FEELINGS, AND ERROR DYNAMICS

The homeostatic perspective has also been invoked in thinking about how the PP framework may be extended to discussions about feelings, emotions and moods (Ainley, Apps, Fotopoulou, & Tsakiris, 2016; Allen et al., 2016; Apps & Tsakiris, 2014; Barrett & Simmons, 2015; Bruineberg & Rietveld, 2014; Clark, 2016; Gu, Hof, Friston, & Fan, 2013; Kiverstien et al., 2020; Seth, 2013, 2014; Seth, Suzuki, & Critchley, 2012). Anil Seth, for example, proposes that PP applies just as neatly to interoception as it does exteroception (2013). Interoception refers to our internal sense of physiological changes in the body—somatic, visceral, vascular, and motor (Craig, 2002, 2003). According to Seth, top-down predictions about the source of interoceptive signals counter-flow with bottom-up interoceptive prediction errors (just as in exteroceptive experiences). Feelings arise from the ongoing integration of these various predictive representations.

Inward-looking error signals are hypothesized to be minimized in an analogous way to sensorimotor predictions: either the error modifies the model to fit the inner world, or autonomic reflexes are initiated which influence the body to fit the prediction. In this model autonomic reflexes are called on to fulfill interoceptive predictions. Consider the example of hunger: when blood sugar levels drop below expected levels interoceptive error is generated. These errors update top-down expectations leading to the subjective experiences of hunger. While the interoceptive error is explained away at one level of the hierarchy as “hunger,” the hunger state itself produces error that is resolved via

autonomic feedback loops that metabolize fats and/or feedback loops that establish and guide allostatic action sequences that lead to finding and consuming sugary foods.

This predictive approach to explaining bodily feelings suggests new possible dimensions to classical somatic theories of emotion. For example, William James defended a view of emotion as the perception of the physiological changes that result from an exciting encounter (1884). “Fear,” for James, was thought to be constructed by our interoceptive perceptions of the internal bodily changes that are characteristic of fear (e.g., sweating, intercostal tightening, etc.). Subjectively speaking, interoceptive awareness manifests as a diverse set of feelings including those of “pain, temperature, itch, sensual touch, muscular and visceral sensations...hunger, thirst, and air hunger” (Craig, 2003, p. 500). The feeling of fear, if James is right, is thus essentially the detection of an interoceptive physiological signature that has already been induced by exposure to the threatening situation. In other words, we do not shake because we are afraid of the angry dog, we *shake* and that shaking makes up our fear.

The trouble with such a simple story is that it suggests a one-to-one mapping between distinct emotional states and distinctive “brute-physiological” signatures, and no such mappings have been found. (Critchley, Wiens, Rotshtein, Öhman, & Dolan, 2004). To remedy this, Seth (2013) and Pezzulo (2014) suggest we may integrate basic information (e.g., about bodily arousal) with higher-level predictions of probable causes. This would help explain why the sensation of a fast-beating heart feels so different when we suspect heart-failure than it does after (say) a bout of vigorous exercise. It also helps explain the kinds of vicious cycle seen in panic attacks, when the best-guess (heart attack) itself impacts bodily response, delivering spurious “evidence” in favor of the heart-attack guess itself.

PP offers an elegant and believable neurocomputational explanation of how these recursive processes might unfold, fueled by the drive to continually reduce the error between top-down predictions and bottom-up interoceptive signals (see also Barrett & Simmons, 2015; Clark, 2016; Seth & Friston, 2016). As Seth notes, these two directions of interaction—from prediction to bodily changes, and from bodily changes to prediction—unfolds “continuously and simultaneously underlining a deep continuity between perception and action” (Seth, 2013, p. 566). The brain’s best guess at the cause of some perceptual signal activates somatic patterns (see also Damasio, 1994), that prepare the body to respond appropriately *and* help the system predict what will happen next. The reactivation includes both autonomic changes and explicit actions (including gestures, facial expressions, postural changes, etc.). These bodily changes provide the basis for the next wave of interoceptive information to be integrated and matched against the evolving prediction (which includes exteroceptive information, memories and predictions), and it is these ongoing reciprocal exchanges that structure our emotional experience. Prediction and incoming signals co-evolve in cycles attempting to minimize discrepancies between model and signal, becoming accessible to conscious awareness as a best-guess stabilizes (Craig, 2002, 2009; Harrison, Gray, Gianaros, & Critchley, 2010).

This provides a very natural way of accommodating large and long-standing bodies of experimental results showing that the character of our experience depends both on the interoception of brute bodily signals and higher-level “cognitive appraisals” (for a review, see Critchley & Harrison, 2013). But the emerging predictive processing account of qualitative experience should not be thought of as a “two-factor” theory as such, but rather, the claim is that a single, highly flexible, inferential process fluidly and constantly combines top-down predictions with bottom-up sensory information. Subjective feeling states are then determined by the ongoing unfolding of this single process.

More recently a view of interoception has begun to take shape that depicts bodily feeling as rather more than just another stream of information for the predictive brain. The idea is that affective changes in the organism’s body might play a special, crucial role in the predictive process itself, through a tight relationship between affectivity and precision weighing.

To bring out this important relationship, we might consider recent work on *error dynamics* (see Hesp et al., 2019; Kiverstein et al., 2017, 2020; Miller et al., 2020). “Error dynamics” refers to the temporal comparison of error reduction rates (see Joffily & Coricelli, 2013; Van de Cruys, 2017). For a predictive organism to thrive it needs to be sensitive to error and error reduction, but also the *rate* at which errors are being reduced. If we think about error minimization happening at a certain speed over time, then the predictive organism must also be sensitive to changes in velocity—accelerations and decelerations in error minimization over time. This sensitivity indicates how well or poorly the organism is doing at reducing uncertainty. Error dynamics, as such, are second-order processes closely related to precision weighting.² In essence, precision is about the error bars on the current best-guesses, while information about the rate of error minimization (relative to expectations) is part of the mechanism that helps set these error bars, and that identifies actions and environments that look set to reward further exploitation. The positive and negative affective tone that accompanies experience may be the conscious reflections of these error dynamics.

Error dynamics, in other words, are made available to the system as emotional valence—feelings of pleasure and displeasure, or states of attraction or repulsion (Joffily & Coricelli, 2013; Van de Cruys, 2017).³ When things are going well for the predictive agent (and its behaviors are resulting in a more certain future) it feels good. When it is struggling to get a handle on the scene, or is unable to manage the complexity of some task, it feels bad. To be clear, this feeling should not be seen as something over and above the tracking of error dynamics, but rather the feelings are a reflection of the quality of the organism's engagement with (or “grip upon”) the environment (see also Polani, 2009). As Van de Cruys (2017) writes, “Emotions ... appear as the continuous non-conceptual feedback on evolving —increasing or decreasing— uncertainties relative to our predictions. The upshot of this view is that the various emotions, from ‘basic’ ones to the non-typical ones such as humour, curiosity and aesthetic affects, can be shown to follow a single underlying logic.”

There is then a conceptual connection between our sensitivity to our own error dynamics, feelings, and the “tendency towards an optimal grip.” Feelings reflect the need to keep expected uncertainty to a minimum in our interactions with the environment. This means that the organism succeeds in improving its overall condition in relation to its environment, keeping in touch with what matters, in ways that reflect both bodily state and trajectories of error resolution. This folds embodied, environmentally embedded value back into the heart of the predictive system.⁴

6 | EXTENDED PREDICTIVE MINDS?

Let us end by seeing how this all scales up to whole brain–body–world ecologies. For here too, the predictive brain plays a crucial and distinctive role. Brains like this will congenitally trade real world action against on-board computation (by using an app, for example) whenever that is estimated to be the best way to reduce key uncertainties (hence to minimize future errors) given current goals. Such “epistemic” actions are chosen, we saw, so as to minimize expected future prediction error. The point to notice is that epistemic actions selected to reduce expected uncertainty in this way can exploit any amount of reliable environmental structure and scaffolding. Inner (brain-based) strategies will thus emerge that rely heavily on the use of external structure and resources.

PP here solves, at least in principle, the so-called “recruitment puzzle” concerning the “extended mind” (Clark, 2008). The puzzle concerns just how the canny cognizer manages to recruit, on the spot, whatever mix of problem-solving resources will yield an acceptable result with a minimum of effort. In Clark's, 2008 treatment, the puzzle was described like this:

“[our story] bequeaths a brand new set of puzzles. It invokes an ill-understood process of ‘recruitment’ that soft-assembles a problem solving whole from a candidate pool that may include neural storage and processing routines, perceptual and motoric routines, external storage and operations, and a variety of self-stimulating cycles involving self-produced material scaffolding [e.g. sketching]. And at its most radical, it depicts that process as proceeding without the benefit of a central controller.” (p. 137)

By minimizing estimated future prediction error, a PP system speaks to all these demands. The “process of recruitment” is simply the forward-looking use of estimated uncertainty (future prediction error) to select transient coalitions of internal and bio-external resources. In this selection process the enabling of internal resources (different brain areas, and different information from within the generative model) is accomplished by the very same means as the selection or enabling of external resources. The changing web of internal neuronal influence is selected using the very same means (variable precision weighting) and for the very same reasons (the reduction of estimated uncertainty) as the use of bio-external resources such as apps and notepads. The guiding principle binding all the cases is simply the energy-efficient minimization of expected future prediction error. And in place of a central controller, we now find only that rolling process of error minimization itself.

The wider world ends up looking somewhat different too. For we can now see much of the human-built world as itself a prime reservoir both of achieved precision estimations and of tools for cheaply estimating precisions on-the-fly. Think, for example, of the way we paint red lines beside roads, or fly red flags on dangerous beaches. These otherwise arbitrary structures attract attention and act as local proxies for precision—for our precision estimating brains (see Roepstorff, 2013). And as we behave in our present niche, we gradually alter it—the roads leading to that red flag flying beach may fall into disuse over time, and new ones emerge leading to safer swimming spots. In the ensuing dance between predictive brains and the forces of cultural and socio-technological change may lie the explanation for much that is distinctive about the human mind.

7 | CONCLUSION

Predictive brains, we have argued, are beautifully positioned to weave thoughts, emotions, mental actions, bodily actions, and environmental opportunities into seamless webs that both serve and express our purposes. In this piece, we have displayed (Table 1) some of the recent evolution of these ideas. Simple appeals to generative models, error minimizing strategies, and precision weighting get us a long way in making sense of what drives the neuroeconomy. But a complete picture of human cognition will require us to “wild the predictive brain” by factoring in the many ways that the predictive brain operates as part of a wider system. This means locating the neural prediction machinery within an environmentally situated, active body, and appreciating the crucial place of affect in the predictive mind.

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CONFLICT OF INTEREST

The authors have declared no conflicts of interest for this article.

AUTHOR CONTRIBUTIONS

Andy Clark: Conceptualization; funding acquisition; project administration; writing-original draft; writing-review and editing. **George Deane:** Conceptualization; writing-original draft; writing-review and editing. **Mark Miller:** Conceptualization; project administration; writing-original draft; writing-review and editing. **Kathryn Nave:** Conceptualization; writing-original draft; writing-review and editing.

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ENDNOTES

- ¹ This picture is itself sometimes grounded in the “free energy principle”: a presumed information-theoretic imperative thought to further explain and motivate the kinds of prediction-based regime examined here. We do not address the free energy principle in this review, but see Friston (2009, 2010)
- ² For more discussion on this point see Kiverstein et al., 2017 and Miller et al., 2020.
- ³ This notion of valence as emerging as a form of “prediction error dynamics” has already found a home in both artificial intelligence and robotics circles (Kaplan & Oudeyer, 2007; Schmidhuber, 2010), and research on intrinsic rewards and adaptive behaviors in humans and non-humans (Kaplan & Oudeyer, 2007).
- ⁴ Addiction then emerges as a case in which this tendency towards an optimal grip breaks down (Miller et al. 2020).

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FURTHER READING

- Anderson, M. L. (2017). Of Bayes and Bullets: An embodied, situated, targeting-based account of predictive processing. In T. Metzinger & W. Wiese (Eds.), *Philosophy and predictive processing*: 3 (pp. 60–73). Frankfurt am Main, Germany: MIND Group.
- Bruineberg, J. (2017). Active inference and the primacy of the ‘I can’. In T. Metzinger & W. Wiese (Eds.), *Philosophy and predictive processing*. Frankfurt am Main, Germany: MIND Group.
- Chemero, A. (2011). *Radical embodied cognitive science*. MIT press, Cambridge.
- Churchland, P. S., Ramachandran, V. S., & Sejnowski, T. J. (1994). A critique of pure vision. In *Large-scale neuronal theories of the brain* (p. 23). Cambridge, MA: MIT Press.
- Clark, A. (2017). Busting out: Predictive brains, embodied minds, and the puzzle of the evidentiary veil. *Noûs*, 51(4), 727–753.
- Clark, A., & Chalmers, D. (1998). The extended mind. *Analysis*, 58(1), 7–19.
- Clark, A. (1999). An embodied cognitive science? *Trends in cognitive sciences*, 3(9), 345–351.
- Constant, A., Ramstead, M. J., Veissière, S. P., & Friston, K. (2019). Regimes of expectations: An active inference model of social conformity and human decision making. *Frontiers in psychology*, 10, 679.

- Craig, A. D., & Craig, A. D. (2009). How do you feel – Now? The anterior insula and human awareness. *Nature Reviews Neuroscience*, 10(1), 59–70.
- Craig, A. D., & Craig, A. D. (2009). How do you feel—now? The anterior insula and human awareness. *Nature reviews neuroscience*, 10(1).
- Dayan, P., Hinton, G. E., Neal, R. M., & Zemel, R. S. (1995). The helmholtz machine. *Neural Computation*, 7(5), 889–904.
- Fabry, R. (2017). Transcending the evidentiary boundary: Prediction error minimization, embodied interaction, and explanatory pluralism. *Philosophical Psychology*, 30(4), 1–20.
- Friston, K., FitzGerald, T., Rigoli, F., Schwartenbeck, P., & Pezzulo, G. (2016). Active inference and learning. *Neuroscience & Biobehavioral Reviews*, 68, 862–879.
- Gładziejewski, P. (2015). Explaining mental phenomena with internal representations. A mechanistic perspective. *Studies in Logic, Grammar and Rhetoric*, 40(1), 63–90.
- Hesp, C., Smith, R., Allen, M., Friston, K., & Ramstead, M. (preprint 2019). Deeply felt affect: the emergence of valence in deep active inference. <https://psyarxiv.com/62pfd/>.
- Hohwy, J. (2016). The self-evidencing brain. *Noûs*, 50(2), 259–285.
- James, W. (1983). What is an emotion?[1884]. In *Collected Essays and Reviews*, (34), (pp. 188–205). <https://doi.org/10.1093/mind/os-IX.34.188>.
- Kiefer, A., & Hohwy, J. (2018). Content and misrepresentation in hierarchical generative models. *Synthese*, 195(6), 2387–2415.
- Kirchhoff, M. D., & Robertson, I. (2018). Enactivism and predictive processing: A non-representational view. *Philosophical Explorations*, 21(2), 264–281.
- Kirchhoff, M. D., & Kiverstein, J. (2019). Extended consciousness and predictive processing: A third wave view. Routledge, New York.
- Kiverstein, J., Miller, M., & Rietveld, E. (2020). How mood tunes prediction: A neurophenomenological account of mood and its disturbance in major depression. *Neuroscience of Consciousness*, 2020(1), niaa003.
- Linson A., Clark A., Ramamoorthy S., & Friston K. (2018). The active inference approach to ecological perception: General information dynamics for natural and artificial embodied cognition. *Frontiers in Robotics and AI*, 5. <https://doi.org/10.3389/frobt.2018.00021>.
- Miller, M., Kiverstein, J., & Rietveld, E. (2020). Embodying addiction: A predictive processing account. *Brain and cognition*, 138, 105495.
- Orlandi, N. (2014). The innocent eye: Why vision is not a cognitive process. In *Philosophy of Mind*. New York: Oxford University Press.
- Orlandi, N. (2016). Bayesian perception is ecological perception. *Philosophical Topics*, 44(2), 327–352.
- Polani, D. (2009). Information: currency of life? *HFSP journal*, 3(5), 307–316.

REFERENCES

- Ainley, V., Apps, M. A., Fotopoulou, A., & Tsakiris, M. (2016). ‘Bodily precision’: A predictive coding account of individual differences in interoceptive accuracy. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 371(1708), 20160003.
- Allen, M., Fardo, F., Dietz, M. J., Hillebrandt, H., Friston, K. J., Rees, G., & Roepstorff, A. (2016). Anterior insula coordinates hierarchical processing of tactile mismatch responses. *NeuroImage*, 127, 34–43.
- Allen, M., & Friston, K. J. (2018). From cognitivism to autopoiesis: Towards a computational framework for the embodied mind. *Synthese*, 195(6), 2459–2482.
- Allen, M., & Tsakiris, M. (2018). The body as first prior: Interoceptive predictive processing and the primacy. In *The interoceptive mind: From homeostasis to awareness* (p. 27). New York: Oxford University Press.
- Apps, M. A., & Tsakiris, M. (2014). The free-energy self: A predictive coding account of self-recognition. *Neuroscience & Biobehavioral Reviews*, 41, 85–97.
- Barrett, L. F., & Simmons, W. K. (2015). Interoceptive predictions in the brain. *Nature Reviews Neuroscience*, 16(7), 419–429.
- Botvinick, M., & Toussaint, M. (2012). Planning as inference. *Trends in Cognitive Sciences*, 16(10), 485–488.
- Bruineberg, J., Kiverstein, J., & Rietveld, E. (2018). The anticipating brain is not a scientist: The free-energy principle from an ecological-enactive perspective. *Synthese*, 195(6), 2417–2444.
- Bruineberg, J., & Rietveld, E. (2014). Self-organization, free energy minimization, and optimal grip on a field of affordances. *Frontiers in Human Neuroscience*, 8, 599.
- Bubic, A., Von Cramon, D. Y., & Schubotz, R. I. (2010). Prediction, cognition and the brain. *Frontiers in Human Neuroscience*, 4, 25.
- Clark, A. (2008). *Supersizing the mind: Embodiment, action, and cognitive extension*. New York: Oxford University Press.
- Clark, A. (2013). Whatever next? Predictive brains, situated agents, and the future of cognitive science. *Behavioral and Brain Sciences*, 36(3), 181–204.
- Clark, A. (2016). *Surfing uncertainty: Prediction, action, and the embodied mind*. New York: Oxford University Press.
- Colombetti, G. (2014). *The feeling body: Affective science meets the enactive mind*. Cambridge, MA: MIT Press.
- Craig, A. D. (2002). How do you feel? Interoception: The sense of the physiological condition of the body. *Nature Reviews Neuroscience*, 3(8), 655–666.
- Craig, A. D. (2003). Interoception: The sense of the physiological condition of the body. *Current Opinion in Neurobiology*, 13(4), 500–505.
- Critchley, H. D., & Harrison, N. A. (2013). Visceral influences on brain and behavior. *Neuron*, 77(4), 624–638.
- Critchley, H. D., Wiens, S., Rotshtein, P., Öhman, A., & Dolan, R. J. (2004). Neural systems supporting interoceptive awareness. *Nature Neuroscience*, 7(2), 189–195.
- Damasio, A. R. (1994). *Descartes’ error: Emotion, rationality and the human brain*. New York: Putnam Berkley Group.
- Di Paolo, E. A. (2005). Autopoiesis, adaptivity, teleology, agency. *Phenomenology and the Cognitive Sciences*, 4(4), 429–452.
- Fink, P. W., Foo, P. S., & Warren, W. H. (2009). Catching fly balls in virtual reality: A critical test of the outfielder problem. *Journal of Vision*, 9(13), 14–14, 14.8.

- Friston, K. (2009). The free-energy principle: A rough guide to the brain? *Trends in Cognitive Sciences*, 13(7), 293–301.
- Friston, K. (2010). The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127–138.
- Friston, K., Rigoli, F., Ognibene, D., Mathys, C., Fitzgerald, T., & Pezzulo, G. (2015). Active inference and epistemic value. *Cognitive Neuroscience*, 6(4), 187–214.
- Friston, K. J., Lin, M., Frith, C. D., Pezzulo, G., Hobson, J. A., & Ondobaka, S. (2017). Active inference, curiosity and insight. *Neural Computation*, 29(10), 2633–2683.
- Gallagher, S. (2017). *Enactivist interventions: Rethinking the mind*. New York: Oxford University Press.
- Gallagher, S., & Allen, M. (2018). Active inference, enactivism and the hermeneutics of social cognition. *Synthese*, 195(6), 2627–2648.
- Gregory, R. L. (1997). Knowledge in perception and illusion. *Philosophical Transactions of the Royal Society of London. Series B: Biological Sciences*, 352(1358), 1121–1127.
- Gu, X., Hof, P. R., Friston, K. J., & Fan, J. (2013). Anterior insular cortex and emotional awareness. *Journal of Comparative Neurology*, 521(15), 3371–3388.
- Harrison, N. A., Gray, M. A., Gianaros, P. J., & Critchley, H. D. (2010). The embodiment of emotional feelings in the brain. *The Journal of Neuroscience*, 30, 12878–12884.
- Helmholtz, H. L. (1867/1910). *Handbuch der physiologischen Optik*. Leipzig: L. Voss. Reprinted, with extensive commentary. In A. Gullstrand, J. von Kries, & W. Nagel (Eds.), *Handbuch der physiologischen Optik* (3rd ed.). Leipzig: Leipzig, Voss.
- Hohwy, J. (2013). *The predictive mind*. New York: Oxford University Press.
- Joffily, M., & Coricelli, G. (2013). Emotional valence and the free-energy principle. *PLoS Computational Biology*, 9(6), e1003094.
- Kant, I. (1781/1929). *Critique of pure reason*. In N. Kemp Smith (trans.). London: Macmillan.
- Kaplan, F., & Oudeyer, P. Y. (2007). In search of the neural circuits of intrinsic motivation. *Frontiers in Neuroscience*, 1, 17.
- Kaplan, R., & Friston, K. J. (2018). Planning and navigation as active inference. *Biological Cybernetics*, 112(4), 323–343.
- Kirchhoff, M. D. (2018). Autopoiesis, free energy, and the life–mind continuity thesis. *Synthese*, 195(6), 2519–2540.
- Kirsh, D., & Maglio, P. (1994). On distinguishing epistemic from pragmatic action. *Cognitive Science*, 18(4), 513–549.
- Kiverstein, J., Miller, M., & Rietveld, E. (2017). The feeling of grip: Novelty, error dynamics, and the predictive brain. *Synthese*, 196(7), 2847–2869.
- Lee, T. S., & Mumford, D. (2003). Hierarchical Bayesian inference in the visual cortex. *Journal of the Optical Society of America. A, Optics, Image Science, and Vision*, 20(7), 1434–1448.
- Marr, D. (1982). *Vision: A computational approach*. San Francisco, CA: Freeman & Co.
- Norcia, A. (2006). The coffer illusion. In F. Macpherson (Ed.), *The illusions index*. Glasgow: The Illusion Index. Retrieved from <https://www.illusionsindex.org/i/coffer-illusion>
- O'Regan, J. K., & Noë, A. (2001). A sensorimotor account of vision and visual consciousness. *Behavioral and Brain Sciences*, 24(5), 939–973.
- Pavlov, I. P. (1927). *Conditional reflexes: An investigation of the physiological activity of the cerebral cortex*. London: Oxford University Press.
- Pezzulo, G. (2014). Why do you fear the bogeyman? An embodied predictive coding model of perceptual inference. *Cognitive, Affective, & Behavioral Neuroscience*, 14(3), 902–911.
- Rao, R. P., & Ballard, D. H. (1999). Predictive coding in the visual cortex: A functional interpretation of some extra-classical receptive-field effects. *Nature Neuroscience*, 2(1), 79–87.
- Roepstorff, A. (2013). Interactively human: Sharing time, constructing materiality. *Behavioral and Brain Sciences*, 36, 224–225. <https://doi.org/10.1017/S0140525X12002427>
- Schmidhuber, J. (2010). Formal theory of creativity, fun, and intrinsic motivation (1990–2010). *IEEE Transactions on Autonomous Mental Development*, 2, 230–247.
- Seth, A. K. (2013). Interoceptive inference, emotion, and the embodied self. *Trends in Cognitive Sciences*, 17(11), 565–573.
- Seth, A. K. (2014). *The cybernetic Bayesian brain*. Open MIND. Frankfurt am Main, Germany: MIND Group.
- Seth, A. K., & Friston, K. J. (2016). Active interoceptive inference and the emotional brain. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 371(1708), 20160007.
- Seth, A. K., Suzuki, K., & Critchley, H. D. (2012). An interoceptive predictive coding model of conscious presence. *Frontiers in Psychology*, 2, 395.
- Sterling, P., & Laughlin, S. (2015). *Principles of neural design*. Cambridge, England: MIT Press.
- Thompson, E. (2007). *Mind in life: Biology, phenomenology, and the sciences of mind*. Cambridge, MA: Harvard University Press.
- Van de Cruys, S. (2017). *Affective value in the predictive mind*. Frankfurt am Main, Germany: MIND Group.
- Varela, F. J. (1991). Organism: A meshwork of selfless selves. In *Organism and the origins of self* (pp. 79–107). Dordrecht, the Netherlands: Springer.
- Varela, F. J., Thompson, E., & Rosch, E. (2016). *The embodied mind: Cognitive science and human experience*. Massachusetts: MIT Press.
- Walsh, K. S., McGovern, D. P., Clark, A., & O'Connell, R. G. (2020). Evaluating the neurophysiological evidence for predictive processing as a model of perception. *Annals of the new York Academy of Sciences*, 1464, 242–268. <https://doi.org/10.1111/nyas.14321>